

Viridiana Lizardo

BIOLOGIST · MACROECOLOGY AND BIOGEOGRAPHY

Zacatecas, México

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Profile

Biologist and entomologist specialising in macroecology and evolutionary biogeography. My research focuses on the evolution and modelling of the ecological niche, integrating phylogenetic, climatic and spatial data to understand biodiversity distribution patterns and their response to climate change. Dung beetles are my main study system, and I also have experience with multi-taxa studies. I develop statistical analyses in R and geographic information systems. My work combines basic research with applications in conservation and science communication.

Education

PhD in Biological Sciences

Ciudad de México, México

UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO (UNAM)

2021–2025

- Thesis: Phylogenetic conservatism of species in the Mexican Transition Zone (with honours)
- Supervisor: Dr Juan José Morrone
- Committee: Dr Enrique Martínez-Meyer and Dr Federico Escobar-Sarria

MSc in Biological Sciences

Ciudad de México, México

UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO (UNAM)

2013–2015

- Thesis: Potential distribution of Phanaeini (Coleoptera: Scarabaeidae) in Mexico and its possible modification by land-use change
- Supervisors: Dr Federico Escobar-Sarria and Dr Santiago Zaragoza-Caballero
- Committee: Dr Juan José Morrone

BSc in Biology

Ciudad de México, México

FACULTAD DE CIENCIAS, UNAM

2007–2012

- Thesis: Analysis of the dung beetle community of Villa de Cos, Zacatecas, Mexico
- Supervisor: Iván Castellanos-Vargas, MSc

Professional experience

Scientific consultant in ecological and spatial modelling

Ciudad de México, México (remoto)

INSTITUTO DE BIOLOGÍA, UNAM – PROGRAMA DE INVESTIGACIÓN EN CAMBIO CLIMÁTICO (PINCC)

Jul 2025–present

- Project “Capacity building for climate change decision-making and update of Mexico’s Nationally Determined Contributions”
- Ecological niche models of key species groups of Mexican biomes under present and future climate scenarios
- Technical inputs for climate policy decision-making
- Supervisor: Dr Enrique Martínez-Meyer

Research projects and collaborations

Collaboration: biogeographic boundaries and phylogenetic endemism

Argentina (remoto)

UNIVERSIDAD NACIONAL DEL COMAHUE, INIBIOMA–CONICET

2025–present

- Data compilation, design and analysis for the project “Biogeographic boundaries as cradles and museums of phylogenetic endemism: the dual evolutionary role of the South American Transition Zone”
- Supervisor: Dr Adriana Ruggiero

Collaboration: e-Scarab, the electronic Scarabaeoidea fauna of Mexico

Xalapa, Veracruz, México

INSTITUTO DE ECOLOGÍA A.C. (INECOL) Y CONABIO

2024–present

- Checklist and database of the Scarabaeoidea fauna of Mexico
- Coordinator: Dr Mario Favila

Collaboration: diversity of copro-necrophagous beetles

Tlaxcala, México

UNIVERSIDAD AUTÓNOMA DE TLAXCALA Y CENTRO TLAXCALA DE BIOLOGÍA DE LA CONDUCTA

2023–2024

- Species distribution modelling with MaxEnt and spatial analysis of taxonomic, functional and phylogenetic alpha and beta diversity
- Invited by Dr Víctor Moctezuma

Collaboration: PHAN-MX database

Xalapa, Veracruz, México

INSTITUTO DE ECOLOGÍA A.C. (INECOL) Y CONABIO

2013–2015

- Manager of the PHAN-MX database
- Compilation, cleaning and standardisation of species distribution data
- Distribution modelling with GARP
- Coordinators: Dr Federico Escobar-Sarria and Dr Jorge M. Lobo

Research stays

Academic stay: Dung Beetle Sorting Session

Oxford, Reino Unido

MUSEUM OF NATURAL HISTORY, UNIVERSITY OF OXFORD

Oct 2015–Nov 2015

- Counting, sorting and identification of Scarabaeidae from Honduras (Operation Wallacea 2015)
- Coordinator: Dr Thomas J. Creedy

Academic stay: potential distribution of Phanaeini

Xalapa, Veracruz, México

INSTITUTO DE ECOLOGÍA A.C. (INECOL)

2014

- Species distribution modelling with GARP, MaxEnt and R
- Analysis and cleaning of geographic distribution databases
- Supervisor: Dr Octavio Rojas-Soto

Other academic experience

Academic placement

Zacatecas, México

POSGRADO EN FILOSOFÍA E HISTORIA DE LAS IDEAS, UNIVERSIDAD AUTÓNOMA DE ZACATECAS

2019–2020

- Review of MSc students' academic projects, document digitisation and administrative support

Social service internship

Ciudad de México, México

COLECCIÓN NACIONAL DE INSECTOS, INSTITUTO DE BIOLOGÍA, UNAM

2009–2010

- Curatorial work, specimen mounting and maintenance, label data capture and macrophotography
- Support in digitisation and linking data to specimens
- Supervisor: Cristina Mayorga, MSc

Peer-reviewed articles

- **Lizardo, V.**, & Ruggiero, A. (2026). Biogeographic boundaries and angiosperm phylogenetic endemism: Highlighting the evolutionary complexity of the South American Transition Zone. *Journal of Biogeography*. <https://doi.org/10.1111/jbi.70183>
- **Lizardo, V.**, Ruggiero, A., & Morrone, J. J. (2025). Patrones de riqueza de especies y conservadurismo filogenético del nicho ecológico en la Zona de Transición Mexicana: evidencia y herramientas para su estudio. *Revista Mexicana de Biodiversidad*, 96, e965347. <https://doi.org/10.22201/ib.20078706e.2025.96.5347>
- **Lizardo, V.**, Escobar, F., Martínez-Meyer, E., & Morrone, J. J. (2024). Adaptive shifts in Phanaeini dung beetles

of the Mexican plateau cenocron in the Mexican transition zone. *Zoologica Scripta*. <https://doi.org/10.1111/zsc.12656>

- Moctezuma, V., **Lizardo, V.**, Arias-Del Razo, I., & Ramírez-Ponce, A. (2024). Overcoming the Wallacean shortfall in sky-islands of central Mexico: the case of copro-necrophagous beetles and two national parks. *Journal of Insect Conservation*. <https://doi.org/10.1007/s10841-024-00598-9>
- **Lizardo, V.**, García Trejo, E. A., & Morrone, J. J. (2024). Niche conservatism and convergence in birds of three cenocrons in the Mexican Transition Zone. *PeerJ*, 12, e16664. <https://doi.org/10.7717/peerj.16664>
- Moctezuma, V., Hernández, B., Sánchez-Huerta, J. L., **Lizardo, V.**, Quiroz-Rocha, G. A., & Navarrete-Heredia, J. L. (2023). Two new species from the Mexican Pacific Slope and new distributional records of *Onthophagus Latreille*, 1802 (Coleoptera, Scarabaeidae). *Journal of Asia-Pacific Entomology*, 26(1), 102015. <https://doi.org/10.1016/j.aspen.2022.102015>
- **Lizardo, V.**, Moctezuma, V., & Escobar, F. (2022). Distribution, regionalization, and diversity of the dung beetle genus *Phanaeus* MacLeay (Coleoptera: Scarabaeidae) using species distribution models. *Zootaxa*, 5213(5), 546–568. <https://doi.org/10.11646/zootaxa.5213.5.4>
- Moctezuma, V., Halffter, G., & **Lizardo, V.** (2021). The *Phanaeus tridens* species group (Coleoptera: Scarabaeoidea): a dung beetle group with genital morphological stasis but a changing ecological niche. *Acta Entomologica Musei Nationalis Pragae*, 447–482. <https://doi.org/10.37520/aemnp.2021.025>
- **Lizardo, V.**, Escobar, F., & Rojas-Soto, O. (2017). Diversity and distribution of *Phanaeini* (Coleoptera: Scarabaeidae: Scarabaeinae) in Mexico. *Zootaxa*, 4358(2), 271. <https://doi.org/10.11646/zootaxa.4358.2.3>
- **Lizardo, V.**, & Castellanos-Vargas, I. (2016). Dung beetle community response to vegetation type and season in an arid zone of the Mexican Plateau. *Southwestern Entomologist*, 41(2), 441–452. <https://doi.org/10.3958/059.041.0215>

Technical reports

- Biodiversidad y cambio climático: estructura, función y contribuciones de la naturaleza a las personas (2026). *Universidad Nacional Autónoma de México (UNAM)*.
- Nature-based solutions in the Country Climate and Development Report for Mexico (2026). *World Bank Group*.
- Segundo informe de actividades. Grupo de Ecosistemas y Biodiversidad (2025). *Universidad Nacional Autónoma de México (UNAM)*.

Project reports

- Informe del proyecto estratégico institucional e-Scarab: la Scarabaeoidea-fauna electrónica de México (2024). *Instituto de Ecología A.C. (INECOL)*.
- Primer informe del proyecto estratégico institucional e-Scarab: la Scarabaeoidea-fauna electrónica de México (2023). *Instituto de Ecología A.C. (INECOL)*.

Books

- **Lizardo, V.** (2023). Un librito de bichos: introducción para niños y niñas a la diversidad entomológica de la región del altiplano. *Amonite*. ISBN 978-607-99717-4-8

Popular science articles

- López Pérez, S., Vega Badillo, V., Núñez Monroy, A., Cifuentes Ruiz, P., **Lizardo, V.**, & González Ramírez, M. (2023). REVISAR TÍTULO COMPLETO ... mexicanas: nombrando, describiendo y clasificando la biodiversidad. *Nextia*.

Intellectual property

Copyright: El libro zacatecano de los bichos

INSTITUTO NACIONAL DEL DERECHO DE AUTOR (INDAUTOR)

- Application 03202306261330270001 (granted)
- Book on the entomology of the Zacatecas highlands for children

México

2023

Teaching

Geographic Information Systems

UNIDAD ACADÉMICA DE AGRONOMÍA, UNIVERSIDAD AUTÓNOMA DE ZACATECAS

- Undergraduate
- Designed and taught a practical course on QGIS and R as a GIS for four groups of Agronomy and Animal Science Engineering (also Jan–Feb and Apr–May 2024)

Zacatecas, México

Feb 2025–Jul 2025

Workshop: New Mexican School curriculum framework, planning, assessment and science didactics

UNIVERSIDAD AUTÓNOMA DE ZACATECAS (UAP-UAZ)

- High-school teachers
- 40 hours

Zacatecas, México

Jan 2025

Tools for macroecological analysis

FACULTAD DE ESTUDIOS SUPERIORES IZTACALA, UNAM

- BSc in Ecology
- Intensive course (6 h) on community ecology methods in R

Estado de México, México

Oct 2024

Vertebrates

CENTRO DE ACTUALIZACIÓN DEL MAGISTERIO EN ZACATECAS

- BSc in Biology Teaching, 7th semester
- Lecture course

Zacatecas, México

Aug 2024–Jan 2025

Multicellular autotrophic organisms

CENTRO DE ACTUALIZACIÓN DEL MAGISTERIO EN ZACATECAS

- BSc in Biology Teaching, 3rd semester
- Lecture course

Zacatecas, México

Aug 2024–Jan 2025

Biostatistics in R

POSGRADO EN CIENCIAS BIOLÓGICAS, UNIVERSIDAD AUTÓNOMA DE TLAXCALA

- MSc (SNP programme)

Tlaxcala, México

Mar 2023–Jun 2023

Thesis seminar

UNIDAD ACADÉMICA DE CIENCIAS BIOLÓGICAS, UNIVERSIDAD AUTÓNOMA DE ZACATECAS

- Undergraduate

Zacatecas, México

Jan 2021–May 2021

Tutoring

Academic tutoring

CENTRO DE ACTUALIZACIÓN DEL MAGISTERIO EN ZACATECAS

- BSc in Biology Teaching
- 40 hours

Zacatecas, México

2024

Thesis supervision and review

Felinos de Zacatecas y contribución del monitoreo comunitario al conocimiento del grupo en el estado, dentro de la comunidad El Chiquihuite, municipio de Susticacán

Zacatecas, México

UNIVERSIDAD AUTÓNOMA DE ZACATECAS

2024–present

- Co-supervisor
- Undergraduate
- Student: Adaya González González
- In progress

Efecto de rasgos corporales sobre la condición fisiológica de *Phanaeus obliquans* y *Phanaeus pseudofurcosus* (Coleoptera: Scarabaeinae)

Ciudad de México, México

UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO (UNAM)

2024–2025

- Reviewer
- Undergraduate
- Student: Gerardo José Martínez Guiscafré
- Completed

Efecto de la estructura y composición del paisaje sobre la diversidad de escarabajos en un gradiente de urbanización en la Ciudad de México

Ciudad de México, México

UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO (UNAM)

2024–2025

- Reviewer
- Undergraduate
- Student: Diana Ivette Fernández González
- Completed

Curriculum design

Workshop “New Mexican School curriculum framework, planning, assessment and science didactics”

Zacatecas, México

UNIVERSIDAD AUTÓNOMA DE ZACATECAS – ESCUELA INTERNACIONAL DE INVIERNO UAZ-SPAUAZ

Jan 2025

- Co-designed and taught the workshop (40 h) for high-school teachers
- Introduction to project- and inquiry-based learning

Peer review

Clave para la determinación de los géneros y subgéneros de escarabajos coprófagos (Coleoptera: Scarabaeidae: Scarabaeinae) de Venezuela

DICTAMEN ACADÉMICO

2024

- Reviewer
- Review of the Spanish-English translation, taxonomic accuracy, biogeographic content and overall scientific rigour

Conference presentations

Distribución potencial de *Phoracanta recurvata* Newman, 1840 (Coleoptera: Cerambycidae), una especie exótica invasiva

Xalapa, Veracruz, México

IV CONGRESO AMXSA

Mar 2025

- Oral presentation

Abordando el déficit Wallaceano en la entomofauna de la provincia Chihuahuense

San Luis Potosí, México

LIX CONGRESO NACIONAL DE ENTOMOLOGÍA

Jun 2024

- Oral presentation

Distribución potencial de las comunidades de escarabajos copronecrófagos de los parques nacionales La Malinche e Iztaccíhuatl-Popocatepetl, con implicaciones para la conservación

Guadalajara, Jalisco, México

XIII REUNIÓN LATINOAMERICANA DE SCARABAEIDOLOGÍA

Oct 2022

- Oral presentation

Conservadurismo de nicho climático de la tribu Phanaeini

Guadalajara, Jalisco, México

XIII REUNIÓN LATINOAMERICANA DE SCARABAEIDOLOGÍA

Oct 2022

- Oral presentation

Niche conservatism in the Mexican transition zone

Sheffield, Reino Unido (en línea)

BRITISH ECOLOGICAL SOCIETY MACROFEST 2022

Jul 2022

- Oral presentation

Patrones geográficos de Phanaeini (Scarabaeidae) en México utilizando modelos de distribución y LDA

San Luis Potosí, México

5.º CONGRESO MEXICANO DE ECOLOGÍA

Apr 2015

- Oral presentation

Biogeografía de Phanaeini en México: inventario de especies, distribución potencial y patrones de distribución

Villa de Leyva, Boyacá, Colombia

X REUNIÓN LATINOAMERICANA DE SCARABAEIDOLOGÍA

Sep 2014

- Oral presentation

Diversidad de escarabajos del estiércol en ambientes xerófilos de Villa de Cos, Zacatecas, México

Boca del Río, Veracruz, México

3.ER CONGRESO MEXICANO DE ECOLOGÍA

Apr 2011

- Oral presentation

Science communication

La dignidad del insecto (podcast)

UNAM

2025

- Co-writer and producer
- Science and humanities podcast

Bicheando como niños (video)

INSTITUTO DE BIOLOGÍA, UNAM

2024

- Video capsule for Children's Day about entomological research

Book launch: Un librito de bichos

Zacatecas, México

CONSEJO ZACATECANO DE CIENCIA, TECNOLOGÍA E INNOVACIÓN – ZIGZAG

2024

- International Day of Women and Girls in Science
- Children's audience

Book launch: Un librito de bichos

Zacatecas, México

CENTRO DE ACTUALIZACIÓN DEL MAGISTERIO EN ZACATECAS

2023

- Talk on scientific language for children
- Student audience

Hilando ciencia (Radio Violeta)

UNIVERSIDAD AUTÓNOMA DEL ESTADO DE HIDALGO

2023

- Radio interview on women in scientific careers

Luciérnaga, science communication programme (SUMA)

UNIVERSIDAD AUTÓNOMA DEL ESTADO DE HIDALGO

Hidalgo, México

2023

- Radio interview about Un librito de bichos

The insect apocalypse

ZIGZAG, CENTRO INTERACTIVO DE CIENCIA Y TECNOLOGÍA DE ZACATECAS

Zacatecas, México

Jul 2022

- Talk on insect population decline
- Exhibition “Bosque de esperanza”

Dung beetles: dirty work for monitoring diversity

COLEGIO DE CIENCIAS Y HUMANIDADES PLANTEL ORIENTE, UNAM

Ciudad de México, México

Mar 2021

- Lecture series “Evolución, sociedad y ambiente” (online)

Biodiversity for children and teenagers

INSTITUTO ZACATECANO DE CULTURA

Zacatecas, México

Oct 2020

- 6th Children’s Art Fandango “Gachita Amador”
- Video series with theory and activities

Mapping dung beetles: a story of biodiversity, mountains, weather, and lazy entomologists

OXFORD UNIVERSITY MEXICAN SOCIETY

Oxford, Reino Unido

Feb 2019

- “Talk About” series

Honours and awards

Doctoral degree with honours

UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO (UNAM)

2025

- Doctoral examination

Theatre programme for children, young people and adults 2025

INSTITUTO NACIONAL DE BELLAS ARTES Y LITERATURA (INBAL)

2025

- Project “La dignidad del insecto: vida y muerte de lo imperceptible”
- Playwrights: Viridiana Lizardo, Nara Hidalgo Pech and Hebzoariba Hernández Gómez

Zacatecas State System of Researchers

CONSEJO ZACATECANO DE CIENCIA, TECNOLOGÍA E INNOVACIÓN (COZCYT)

2024

Scholarships

National Postgraduate Scholarship

CONAHCYT / SECIHTI

2021–2025

- PhD

Timely Graduation Award

UNAM

2015

- MSc

Postgraduate Studies Support Programme (PAEP)

UNAM

2014

- Academic stay at INECOL, Xalapa

National Postgraduate Scholarship

CONACYT

2013–2015

- MSc

Undergraduate Academic Strengthening Programme

UNAM

2010

- Faculty of Sciences

Research interests

- **Macroecology and biogeography:** Detecting and describing patterns; identifying and testing causal processes
- **Ecological niche:** Modelling with different algorithms; ecological and evolutionary theory behind the methods; niche evolution; role in diversity gradients
- **Phylogenetic comparative methods:** Applied to the ecological niche
- **Climate change and biodiversity:** Climate scenarios, niche models and inputs for public policy
- **Dung beetles:** Community ecology; diversity patterns; taxonomy, evolution and biogeography of the tribe Phanaeini

Skills

- **Spatial analysis and GIS:** R (expert: sf, terra, stars); QGIS (advanced); Python (basic)
- **Niche and distribution modelling:** Training, evaluation, hyperparameter tuning and automated batch generation in R (MaxEnt, SDMtune)
- **Biodiversity data management:** Compilation, cleaning, georeferencing, geocoding, taxonomic harmonisation and synonymy, GBIF data mining; SQL (basic)
- **Automation and visualisation:** Reproducible workflows, cartography and data visualisation
- **Scientific communication:** Writing articles in English and Spanish; outreach materials; in-person and remote collaboration

Languages

- **Spanish:** Native
- **English:** Advanced, certified
- **Portuguese:** Intermediate